Requirements

The frontend app (TimberSense) is for the autonomous timber cruising product, which captures lidar, lidar imu, camera and GPS information and processes it to compute DBH, height, species, volume and other parameters from the captured data, where we want to show the DBH, the height, and the species of the trees, and the visualisation of the whole plot in 3D.

- **Need to show an interactive viewer which shows the colored forest point cloud interactively, with each tree hoverable and selectable to view the information about the tree, including the DBH, height and species,**
- **The pointcloud shoule be able to display in RGb colored, segment colored by tree ID, and by semantic label ground, wood, and foliage.**
- **When a tree is hovered over, it should be highlighted; when clicked, it should have the DBH markers and height markers activated, and other derived information like volume, basal area, and all the details listed on the sidebar.**
- **It should have an option to overlay the trajectory that is covered to capture the plot, and an option to trace the trajectory live (as real-time capture) and view the four camera feeds from cameras that are placed at different angles, while the trajectory overlay follows the trajectory where the user was at that particular instance in real time while the camera feed is being played back. You can check the timestamps from imu poses and trajectory and the video frames to do this.**
- **There should be a Map overlay option. As I have GPS coordinates, to show on a global map where this plot is located on the world. And clicking on that plot gets you to the 3d pointcloud map, in an animated way.**
- **The App should have a way to show cinematic flythrough through the pointcloud going through trees, and around the whole plot, with point cloud viewed in centered perspective so that it looks aesthetic to view.**
- **Generally, the pointcloud should be viewed in a centred perspective, like the option in cloud compare. It looks aethetic that way.You can also add a light source, and edl shader like thing etc to make the point cloud look more realistic and beautiful like youre actually standing in the forest between the trees.**
- **The App should have a login screen with (default credentials for now), then on the main screen should be a tab for projects (you can take the idea from other carbon stocking, forestry management and timber cruising apps), and the projects should contain plots, block-wise distributed, and the user should be able to go to each block to analyse. There should also be a project summary and a block-wise summary somewhere. The app should have other navigation buttons that make it feel like a complete forestry management. Timber cruising portal. Other stuff doesn't need to be functional, but the buttons and stuff should be there to give a feel of the complete app. The projects will also be hardcoded right now. There should be 3 4 dummy projects, and only one functional project for the demo, which contains many blocks and plots, of which 2 or 3 plots are actual plots, whose data I am providing for this interactive view and everything.**

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Theme

- **Take the Farmevo logo and theme to the app that is colors and other themes. Website link for aesthetics.**
- **The UX should be very professional and intuitive, and have a good feel.**
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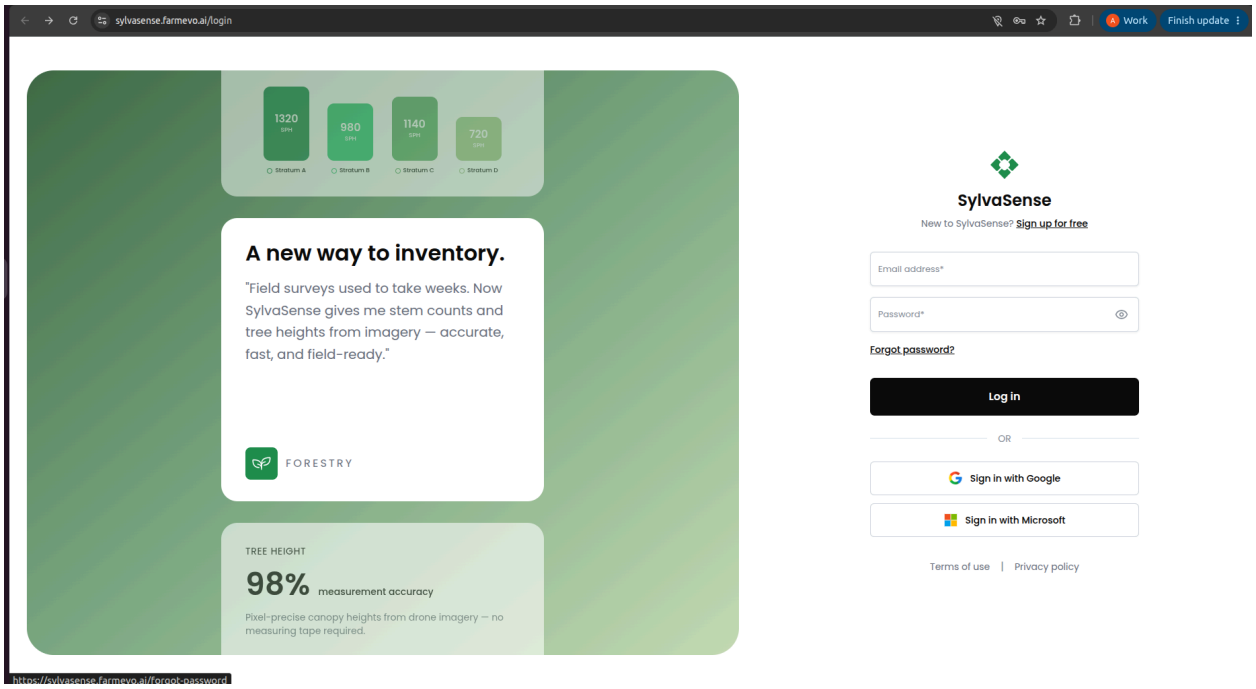
Suggestions

- **Tell me what frontend framework this should be developed on, so that it can work as a browser-based application, same as Sylvasnese or some other.**

- **Generate the whole plan of all the screens and workflow.**

Farmevo website: https://farmevo.ai/
Claude artifact developed previously for a basic idea of the app:
https://claude.ai/artifact/F1dTNL5KD62B2oaPUuBvy3?sk=gc3WDaRjyGkqGRm6XtiP3g

The Ui and feel can be motivated by the Farmevo Sylvssense app. This one should be similar in UX and design to Sylvasense.
https://github.com/Farmevo-Technologies/sylvasense
Here are the snapshots of the Sylvasense app.



sylvasense.farmevo.ai
Work
Finish update
farmevo
Dashboard
Search block
Select Region
Upload Data
48.56 credits
Upgrade
Share
Google Maps
Imagery ©2026 Maxar Technologies

sylvasense.farmevo.ai
Work
Finish update
farmevo
Dashboard
Search block
Southern Interior
Add Block
Upload Data
48.56 credits
Upgrade
Share
Google Maps

sylvasense.farmevo.ai
Work
Finish update
farmevo
Dashboard
Search block
Southern Interior
Add Block
Upload Data
48.56 credits
Upgrade
Share
2 blocks
2 flown · 0 pending
A18667/216/LV4357 (RG)
A18667/947/LV4086 (RG)
Google Maps

sylvasense.farmevo.ai
Work
Finish update
farmevo
Dashboard
Search block
Southern Interior
Add Block
Upload Data
48.56 credits
Upgrade
Share
A18667/947/LV4086 (RG)
A18667/216/LV4357 (RG)
Google Maps
Imagery ©2026 Maxar Technologies

sylvasense.farmevo.ai
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Finish update
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Dashboard
Search block
Southern Interior
Add Block
Upload Data
48.56 credits
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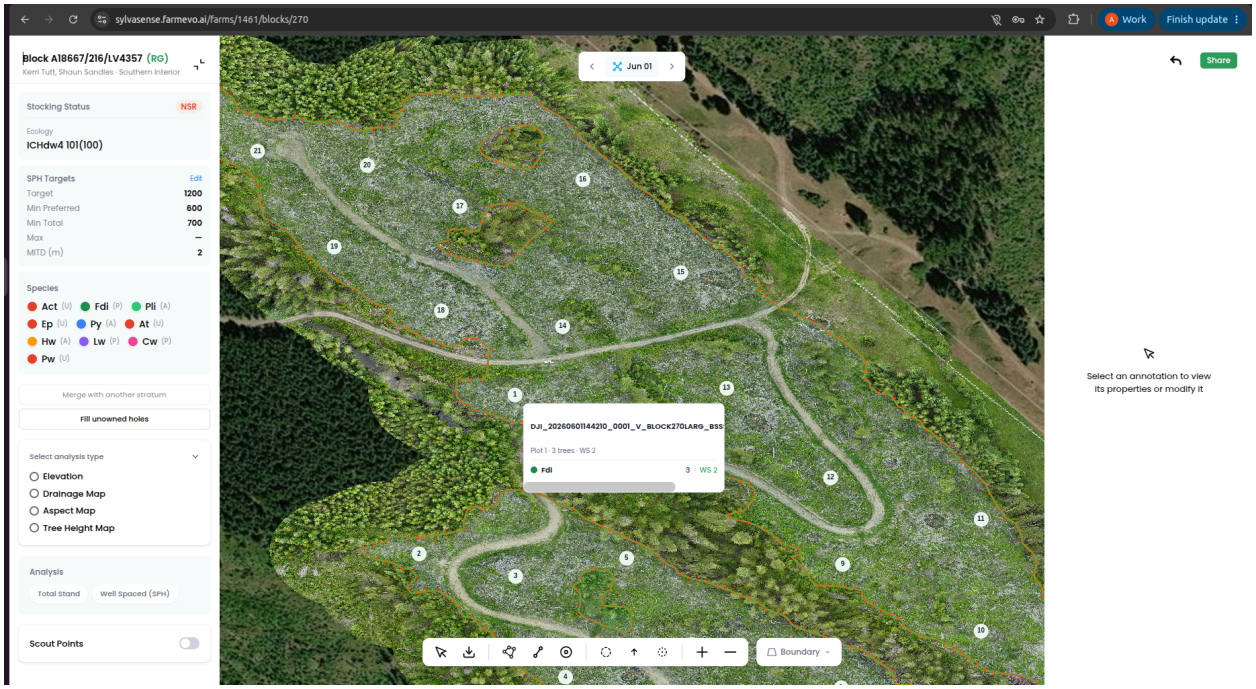
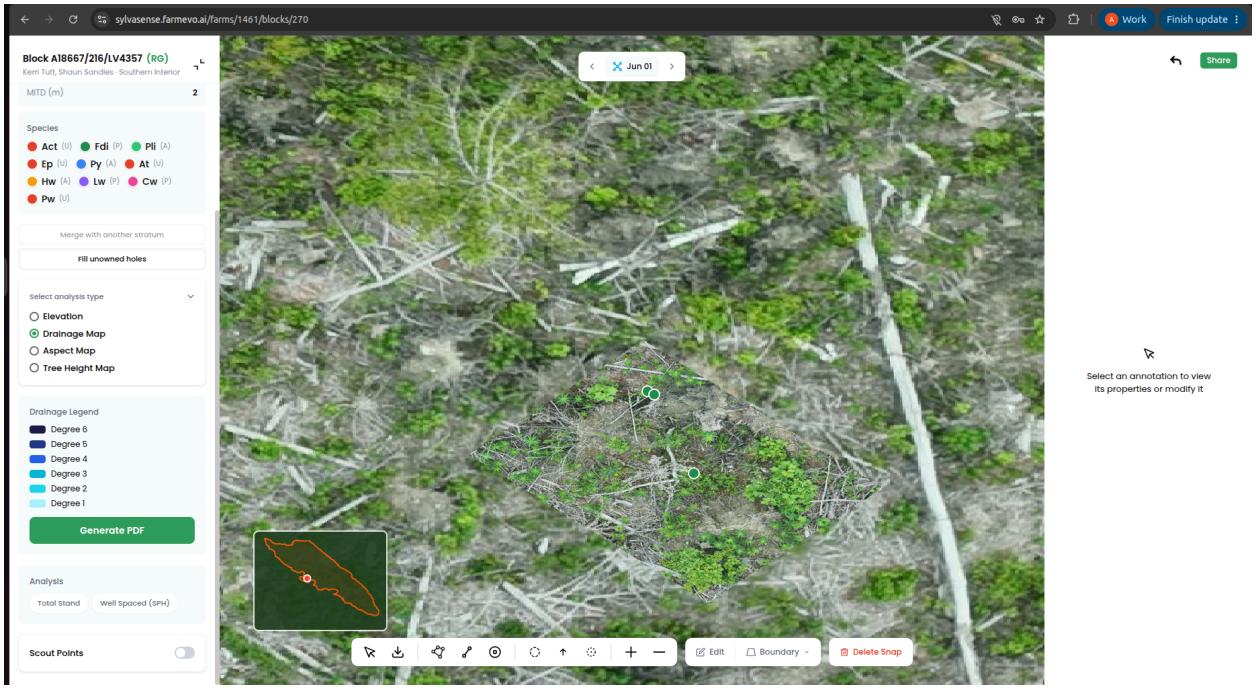
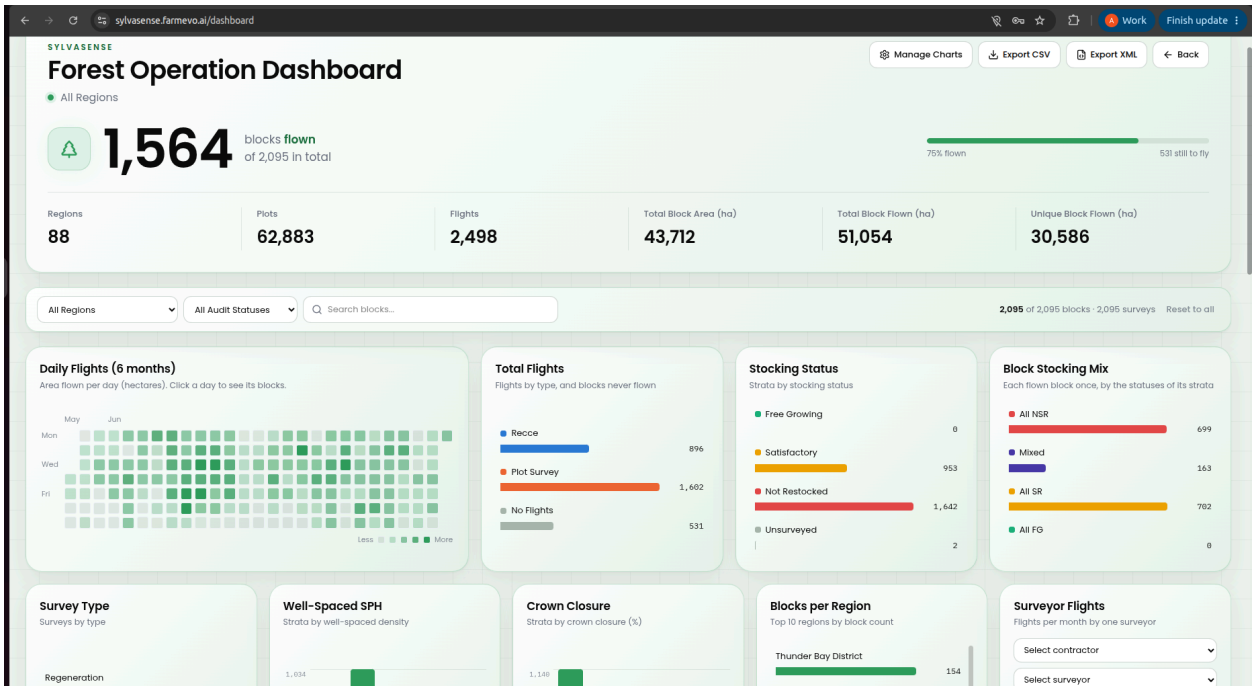
Share
12.60 Ha
Block A18667/216/LV4357 (RG)
Southern Interior
SSW
Strata
Edit Block
Block Summary
Audit Report
Scout Images Report
1
12.60 ha
Ep, Act, Fdi, Hw, Pli, Cw, Lw, Py, Pw, At
1200 target SPH
View Block
A18667/947/LV4086 (RG)
A18667/216/LV4357 (RG)
A18667/216/LV4357 (RG)
Google Maps
Imagery ©2026 Maxar Technologies

sylvasense.farmevo.ai/farms/1461/blocks/270
Work
Finish update
Block A18667/216/LV4357 (RG)
Kerri Tutt, Shaun Sandles · Southern Interior
Stocking Status
NSR
Ecology
ICHdw4 101(100)
SPH Targets
Edit
Target 1200
Min Preferred 600
Min Total 700
Max –
MITD (m) 2
Species
Act (U) Fdi (P) Pli (A)
Ep (U) Py (A) At (U)
Hw (A) Lw (P) Cw (P)
Pw (U)
Merge with another stratum
Fill unowned holes
Select analysis type
Elevation
Drainage Map
Aspect Map
Tree Height Map
Analysis
Total Stand
Well Spaced (SPH)
Scout Points
Jun 01
Share
Select an annotation to view its properties or modify it
Boundary

Block:


sylvasense.farmevo.ai/farms/1461/blocks/270
Work
Finish update
Block A18667/216/LV4357 (RG)
Kerri Tutt, Shaun Sandles · Southern Interior
Stocking Status
NSR
Ecology
ICHdw4 101(100)
SPH Targets
Edit
Target 1200
Min Preferred 600
Min Total 700
Max –
MITD (m) 2
Species
Act (U) Fdi (P) Pli (A)
Ep (U) Py (A) At (U)
Hw (A) Lw (P) Cw (P)
Pw (U)
Merge with another stratum
Fill unowned holes
Select analysis type
Elevation
Drainage Map
Aspect Map
Tree Height Map
Analysis
Total Stand
Well Spaced (SPH)
Scout Points
Jun 01
DJI_20260601144210_0001_V_BLOCK270LARG_BSS
Plot 1 · 3 trees · WS 2
Fdl 3 · WS 2
Boundary
Share
Select an annotation to view its properties or modify it


sylvasense.farmevo.ai/farms/1461/blocks/270
Work
Finish update
Block A18667/216/LV4357 (RG)
Kerri Tutt, Shaun Sandles · Southern Interior
MITD (m) 2
Species
Act (U) Fdi (P) Pli (A)
Ep (U) Py (A) At (U)
Hw (A) Lw (P) Cw (P)
Pw (U)
Merge with another stratum
Fill unowned holes
Select analysis type
Elevation
Drainage Map
Aspect Map
Tree Height Map
Drainage Legend
Degree 6
Degree 5
Degree 4
Degree 3
Degree 2
Degree 1
Generate PDF
Analysis
Total Stand
Well Spaced (SPH)
Scout Points
Jun 01
Edit
Boundary
Delete Snap
Share
Select an annotation to view its properties or modify it


sylvasense.farmevo.ai/dashboard
Work
Finish update
SYLVASENSE
Forest Operation Dashboard
All Regions
Manage Charts
Export CSV
Export XML
Back
1,564
blocks flown
of 2,095 in total
75% flown
531 still to fly
Regions
88
Plots
62,883
Flights
2,498
Total Block Area (ha)
43,712
Total Block Flown (ha)
51,054
Unique Block Flown (ha)
30,586
All Regions
All Audit Statuses
Search blocks...
2,095 of 2,095 blocks · 2,095 surveys
Reset to all
Daily Flights (6 months)
Area flown per day (hectares). Click a day to see its blocks.
May
Jun
Mon
Wed
Fri
Less
More
Total Flights
Flights by type, and blocks never flown
Recce
896
Plot Survey
1,602
No Flights
531
Stocking Status
Strata by stocking status
Free Growing
0
Satisfactory
953
Not Restocked
1,642
Unsurveyed
2
Block Stocking Mix
Each flown block once, by the statuses of its strata
All NSR
699
Mixed
163
All SR
702
All FG
0
Survey Type
Surveys by type
Regeneration
Well-Spaced SPH
Strata by well-spaced density
1,034
Crown Closure
Strata by crown closure (%)
1,148
Blocks per Region
Top 10 regions by block count
Thunder Bay District
154
Surveyor Flights
Flights per month by one surveyor
Select contractor
Select surveyor